



BUSINESS BLUEPRINT

AOP AUTOMATION FOR NMD PE PLANT (LDPE, LLDPE1&2)

Honeywell

Site Name	Reliance Industries Ltd.
Track Name	RIL AOP Automation for NMD LDPE, LLDPE1&2 Plant
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Business Blueprint

AOP Automation for NMD LDPE, LLPE1&2 Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)



Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	01-09-2024	HON	Issued for Review
1.0	16-09-2024	HON	Issued for Review (revised norms logic)
2.0	10-02-2025	HON	Issued for Review (updated RIL comments)
3.0	20-02-2025	HON	Issued for Review (Resolved RIL comments)
4.0	25-04-2025	HON	Issued for Approval
5.0	18-07-2025	HON	Site Specific Changes done and logic updated. Issued for approval
6.0	09-08-2025	HON	Norms Logic and Shutdown allocation logic update.

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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for NMD LDPE, LLDPE1&2 plant for NMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Polyethylene plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple polyethylene (LLDPE, LDPE, HDPE) plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business as the next revised budget by the end of Jan. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for PE technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
O-836243B24R01	R01	AOP Automation for O2C Proposal	 O-836243B24R01 RIL AOP Automation f
PE effective Operating Hours calculation	-	RIL AOP - PE Maintenance Screen – Reliance to Confirm on logic	

NMD PE DCT	-	Digital AOP NMD PE DCT	 NMD%20PE%20DCT %20Rev05.xlsx
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Table 1 Reference Documents

2. AOP Design Basis

The below table captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release post review and discussion

Table 2 AOP Functional Scope

Function-wise detailed design basis is captured in the below sections.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

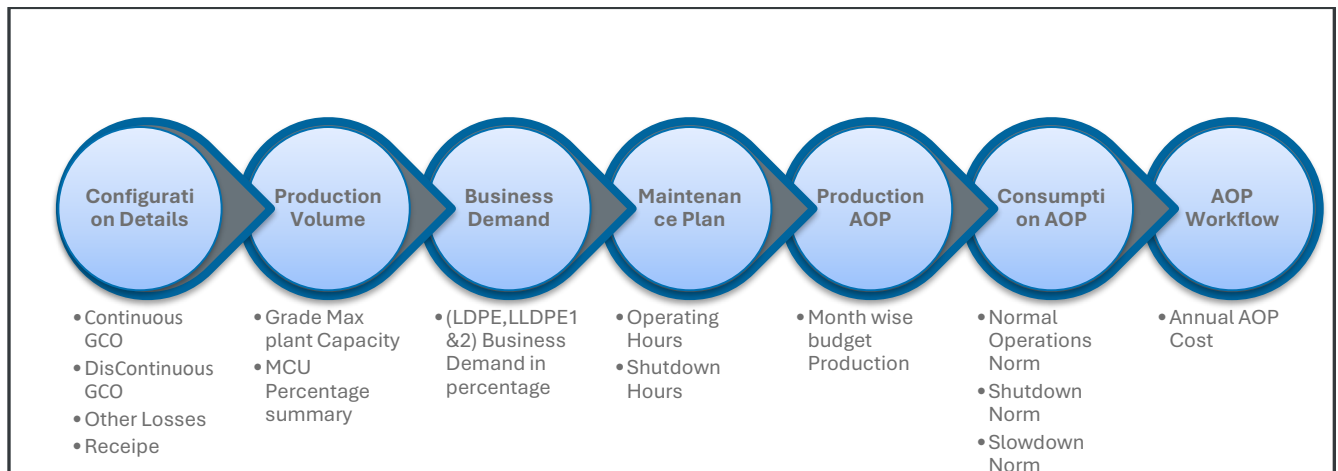


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Configuration

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data are required to calculate Production and consumption. data, Startup losses for raw material & Utilities, other losses, Recipe, Continuous Grade change and discontinuous grade change. There are different tabs available to configure the above parameter under the Configuration screen to support Production and Consumption Norms calculations and reports generation.

By default, the values manual entries entered for the current year will be automatically carried forward to the next year.

In configuration screen, below additional features and tabs are provided.

- **AOP historical period** basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- **Recipe tab** is provided to enter the catalyst chemical recipe value which will not change month wise but can vary grade wise.
- **import and export** options to import and export manual entry parameters from configuration screen is provided.

In addition to the above common tabs there are specific tabs for each unit such as LDPE, LLDPE1 and LLDPE2

LDPE

- **Startup losses** are material wise losses during the startup period.
- **Other losses** are material wise other losses.
- **Shutdown Rate (Per HR)** is rate of losses during shutdown period.

LLDPE1&2

- **Number of Continuous grade change.**
- **Number of Discontinuous grade change.**

Configuration

Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE

Aniket

1 Configuration

2 Shutdown/TA Activities

3 Slowdown Activities

4 Production Volume Data

5 Business Demand

6 Maintenance Details

7 Production AOP

8 Normal Operation Norms

9 Shutdown Norms

10 Slowdown Norms

11 Consumption AOP

12 Feed Stock Availability

PE / NMD / LDPE

AOP Historical Period Basis

Start Date01-04-2024

End Date31-10-2024

Load

(Last refreshed data on: 17-07-2025 10:17 AM for the period from 01-04-2024 to 31-10-2024)

AOP Design Basis

❖ Startup Losses

[illegible]

❖ Other Losses

Configuration Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

Startup Losses Other Losses Shutdown Norms Receipte

Save

Particulars	U...	Apr...	Ma...	Jun...	Jul-25	Au...	Sep...	Oct...	Nov...	Dec...	Jan...	Feb...	Mar...	Re...
Raw Material														
LDPE Ethylene Consumption	MT	10	5.5	5.5	3.5	2.5	1.5	9.5	10.5	11.5	9.5	0	0	Test1
Utility Consumption														
LP Steam_Dis	MT	4.5	4.5	6	9	6	7.5	0	0	0	0	0	0	Test
Power_Dis	KWH	10.5	5	10.5	3	1.5	2.5	3.5	6.5	3.5	10.5	5.5	10	Test
Instrument Air	NM3	2.5	5.5	6.5	9.5	7	3	5.3	2.8	7.6	9.5	9.5	1.6	Test
Nitrogen Gas	NM3	10.5	12	11	6	7.5	6.5	8.5	1.4	1.8	5.5	0	5	Test
Service air	NM3	8.5	2.5	6.5	3.5	5.5	2.5	4.5	9.5	1.5	5.5	3.5	10	Test
Water	M3	2.5	1.5	1.5	1.6	3.5	2.5	5.5	3.5	2.5	1.5	9.5	4.5	Test
HP Steam_Dis	MT	0.5	0.4	0.3	0.8	0.6	5.5	0.935	1.5	1.5	1.6	1.3	1.8	Test
D M Water	M3	1.5	1.5	1.5	2.5	3.5	4.5	5.5	4.5	8.5	1.5	3.5	1.5	Test
Cooling Water 2	M3	0.5	1.5	0.5	0.6	0.9	0.6	0.45	0.5	1.5	1.5	1.5	5.5	Test

❖ Shutdown Norms

Configuration Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

Startup Losses Other Losses Shutdown Norms Receipte

Save

Constant	Particulars	Values	Remark
Raw Material			
Shutdown Norms	LDPE Ethylene Consumption	0	Test
Utility Consumption			
Shutdown Norms	D M Water	1.247	Test
Shutdown Norms	Nitrogen Gas	177.483	Insert
Shutdown Norms	Instrument Air	283.955	Test
Shutdown Norms	Cooling Water 2	2660.471	Test
Shutdown Norms	LP Steam_Dis	6.70225	Test
Shutdown Norms	Service air	6.702	Test
Shutdown Norms	Power_Dis	835.391	Insert
Shutdown Norms	HP Steam_Dis	2.412	Test
Shutdown Norms	Water	1.741	Test

❖ Recipe

Configuration Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities 4 Production Volume Data 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

PE / NMD / LDPE

AOP Historical Period Basis

Start Date: 01-04-2024 End Date: 31-10-2024 Load (Last refreshed data on: 17-07-2025 10:17 AM for the period from 01-04-2024 to 31-10-2024)

AOP Design Basis

Startup Losses Other Losses Shutdown Norms Receipte

Save

Receipte	1020FA20	1070LA17	1003FA20	1005FY20
Oleamide				500

❖ **Continuous grade change**❖ **Discontinuous grade change**

❖ Other Losses

❖ Recipe

Recipe	F46003E	L60075N	LL36RA045	M60075	HD50MA180	HD55EF010
Calcium stearate		1000	1500	1000	1000	2500
Chimassorb-944/UV Stabilizer		2000				
DHT-4A						
DSTDP			1000			
Erucylamide						
Ing1010-Primary AO		500	500	500	500	500
Ing168-Secondary AO		1000	1000	1000	1000	1500
Optiblock						
Optical Brightner		5		5		
PBA						
ZNBT						

Figure 2 Digitized AOP screen for Configuration/manual Inputs

2.1.1.2 Production Volume Data

The Max Capacity (MC) of NMD LDPE, LLDPE1&2 plants is defined on MC values captured in MCU portal. Last value for each grade in selected AOP basis period is selected as MC for that grade.

For example: For grade M60075 and if AOP basis period is selected as April-23 to Oct-23. The last MC value recorded for M60075 in MCU portal that might be 31st Oct 23 will be selected as Basis for next year production calculation by converting it to TPH.

With this logic All grade MC is derived for NMD LDPE and LLDPE 1&2

NOTE- AOP historical period basis added in configuration screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive MC values.

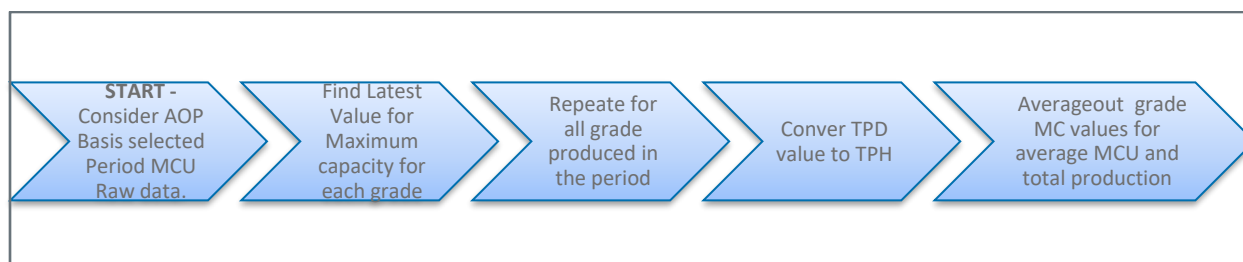


Figure 3 Digitized AOP Steps to derive MC values for NMD LDPE, LLDPE1&2

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary but in case of LDPE and LLDPE1&2 production rate value will be same for all months for each grade.

This information is captured in the AOP digitization portal as seen in the screenshot below.

Production Volume Data Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities 4 **Production Volume Data** 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

PE / NMD / LDPE

Save Select UOM TPH

Parti...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
1005PY20	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	Click to add r...
1020FA20	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	Click to add r...
1070LA17	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	Click to add r...

Percentage Summary

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
1005PY20	100	100	100	100	100	100	100	100	100	100	100	100
1020FA20	100	100	100	100	100	100	100	100	100	100	100	100
1070LA17	100	100	100	100	100	100	100	100	100	100	100	100

Production Volume Data Year: 2025-26 Vertical: PE Site: NMD Plant: LLDPE1 Aniket

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities 4 **Production Volume Data** 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

PE / NMD / LLDPE1

Save Select UOM TPH

Parti...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
HD50MA180	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	Click to add r...
HD53EA010	20	20	20	20	20	20	20	20	20	20	20	20	Click to add r...
HD55EF010	20	20	20	20	20	20	20	20	20	20	20	20	Click to add r...
M60075	21	21	21	21	21	21	21	21	21	21	21	21	Click to add r...

Percentage Summary

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
HD50MA180	100	100	100	100	100	100	100	100	100	100	100	100
HD53EA010	100	100	100	100	100	100	100	100	100	100	100	100
HD55EF010	100	100	100	100	100	100	100	100	100	100	100	100
M60075	100	100	100	100	100	100	100	100	100	100	100	100

Production Volume Data

Year: 2025-26

Vertical: PE

Site: NMD

Plant: LLDPE2

Aniket

PE / NMD / LLDPE2

Save

Select UOM
TPH

Parti...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
B5010	14	14	14	14	14	14	14	14	14	14	14	14	Click to add r...
B5502	17	17	17	17	17	17	17	17	17	17	17	17	Click to add r...
HD53EA010	15.917	15.917	15.917	15.917	15.917	15.917	15.917	15.917	15.917	15.917	15.917	15.917	Click to add r...
HD55EF010	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	Click to add r...
LL20FA010	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	Click to add r...
LL36RA045	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	Click to add r...
LL36RA045UV	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	Click to add r...
P4609	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	Click to add r...

Percentage Summary

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
B5010	100	100	100	100	100	100	100	100	100	100	100	100	100
B5502	100	100	100	100	100	100	100	100	100	100	100	100	100

Figure 4 Digitized AOP screen for Production Volume data

The '**Calculate**' button on this screen allows the user to compute the MC value if any manual changes have been made to the AOP basis selection on the configuration screen

2.1.1.3 Business Demand

The grade mix demand will be provided by the Business team and added in AOP portal manually. In the AOP digitization application, Percent monthly demand for each individual grade will be recorded.

An example of the same is mentioned in the table below.

Business Demand

Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

Production Volume Data (MT) (This is a reference for entering the Business Demand value)

Select UOM
TPH

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
1005FY20	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708	13.708
1020FA20	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75	13.75
1070LA17	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833	12.833

Business Demand Data

Save

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Rem...
Production													
1005FY20	30	40	50	60	70	50	20	45	40	30	60	10	Demand upd...
1020FA20	50	30	40	20	10	30	20	25	60	30	20	10	Updated...
1070LA17	20	30	10	20	20	20	60	30	0	40	20	80	Demand Upd...

Business Demand

Year: 2025-26 Vertical: PE Site: NMD Plant: LLDPE1 Aniket

Select UOM
TPH

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
HD50MA180	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25	22.25
HD53EA010	20	20	20	20	20	20	20	20	20	20	20	20
HD55EF010	20	20	20	20	20	20	20	20	20	20	20	20
M60075	21	21	21	21	21	21	21	21	21	21	21	21

Business Demand Data

Save

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Rem...
Production													
HD50MA180	10	20	20	20	20	20	20	10	20	20	10	20	Updated...
HD53EA010	15	10	20	20	20	20	20	20	10	20	10	20	Updated...
HD55EF010	10	10	10	30	10	30	10	30	30	30	10	10	Updated...
M60075	20	20	10	10	20	60	10	20	10	10	20	10	Updated...

Business Demand

Year: 2025-26 Vertical: PE Site: NMD Plant: LLDPE2 Aniket

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
LL36RA045	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5
LL36RA045UV	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5	18.5
P4609	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5

Business Demand Data

Save

Partic...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Rem...
Production													
B5010	10	10	10	10	10	10	10	10	10	10	10	10	Test
B5502	20	20	20	20	20	20	20	20	20	20	20	20	Test
HD53EA010	30	30	30	30	30	30	30	30	30	30	30	30	Test
HD55EF010	40	40	40	40	40	40	40	40	40	40	40	40	Test
LL20FA010	50	50	50	50	50	50	50	50	50	50	50	50	Test
LL36RA045	40	40	40	40	40	40	40	40	40	40	40	40	Test
LL36RA045UV	50	50	50	50	50	50	50	50	50	50	50	50	Test
P4609	50	50	50	50	50	50	50	50	50	50	50	50	Test

Figure 5 Grade-wise Business demand

2.1.1.4 Maintenance/Shutdown

The operations team will provide the shutdown and slowdown plan to calculate month-wise operating hours available for production.

The screenshot below depicts the shutdown activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration (hrs) to calculate the SD-To date/time.

Shutdown/TA Activities

Year: 2025-26 Vertical: PE Site: NMD Plant: LDPE Aniket

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities 4 Production Volume Data 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

PE / NMD / LDPE

Add Item Save

Shutdown Desc	SD - From	SD - To	Duration (hrs)	Shutdown Basis	Action
Shutdown	02-04-2025 12:00 AM	03-04-2025 12:00 AM		24:00 Planned Shutdown Updated.	

Figure 6 Digitized AOP screen for Shutdown activities

The slowdown activities can be calculated based on the slowdown rate and hours. This is called Basis Shutdown, equivalent of Slowdown.

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate} * \text{Slowdown hrs}) / \text{MCU}$$

The below screen depicts slowdown activities planned as will be captured in the digitized platform.

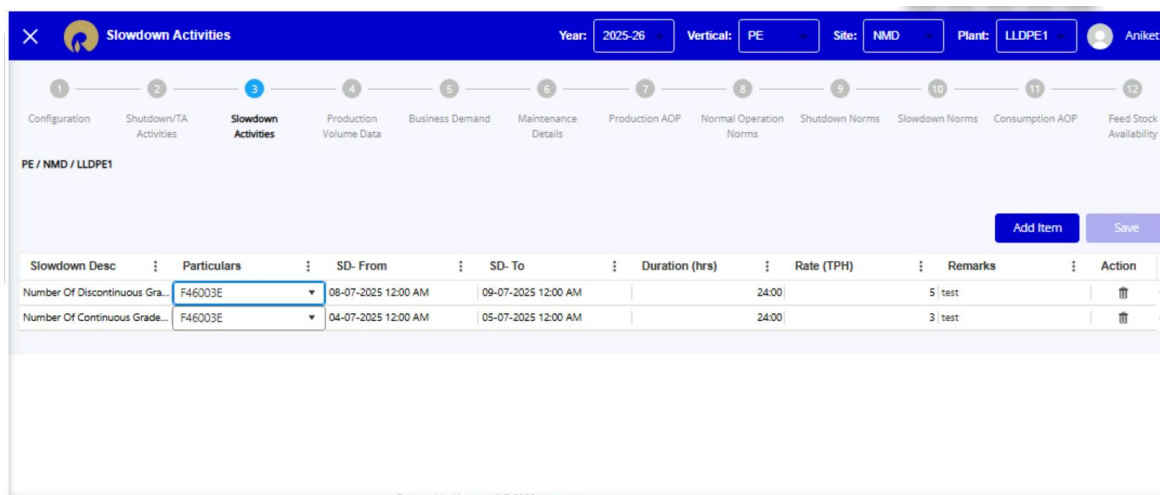


Figure 7 Digitized AOP screen for Slowdown Activities

Using this number, the operating hours available in a month will be calculated as -

$$\text{Number of Operating Hours in a month} = \text{Total Number of hrs in a month} - (\text{Shutdown Hrs} + \text{Slowdown converted to equivalent SD Hrs})$$

The screenshot below captures an example of the number of operating hours, Shutdown hours, and effective slowdown loads available in a month wise.

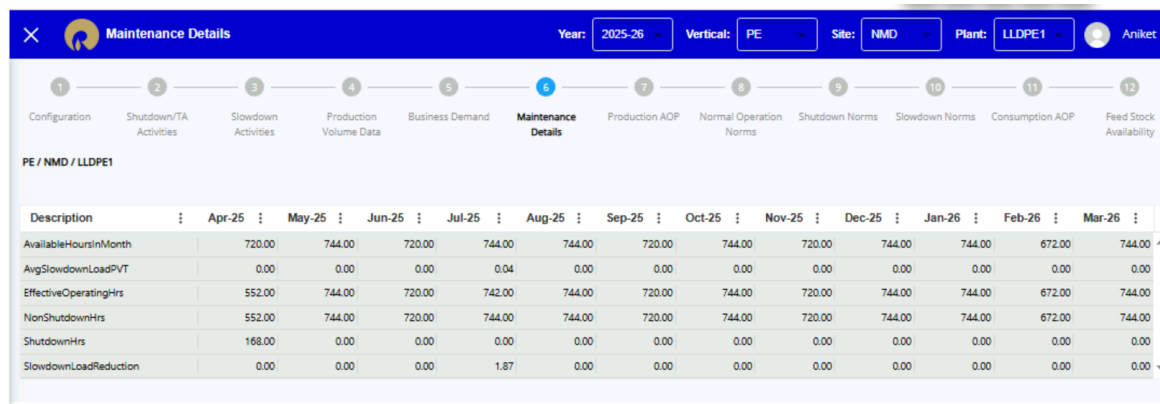


Figure 8 Number of operating hours

Note: RIL to validate slowdown and shut down hours calculated in System and raise any exception observed.

2.1.1.5 Calendar Year Based Production Calculations

Using the Production Volume data, business demand, and maintenance plan, the month-wise and grade-wise budgeted production quantity will be derived.

The calculation logic is as below.

For a single-grade

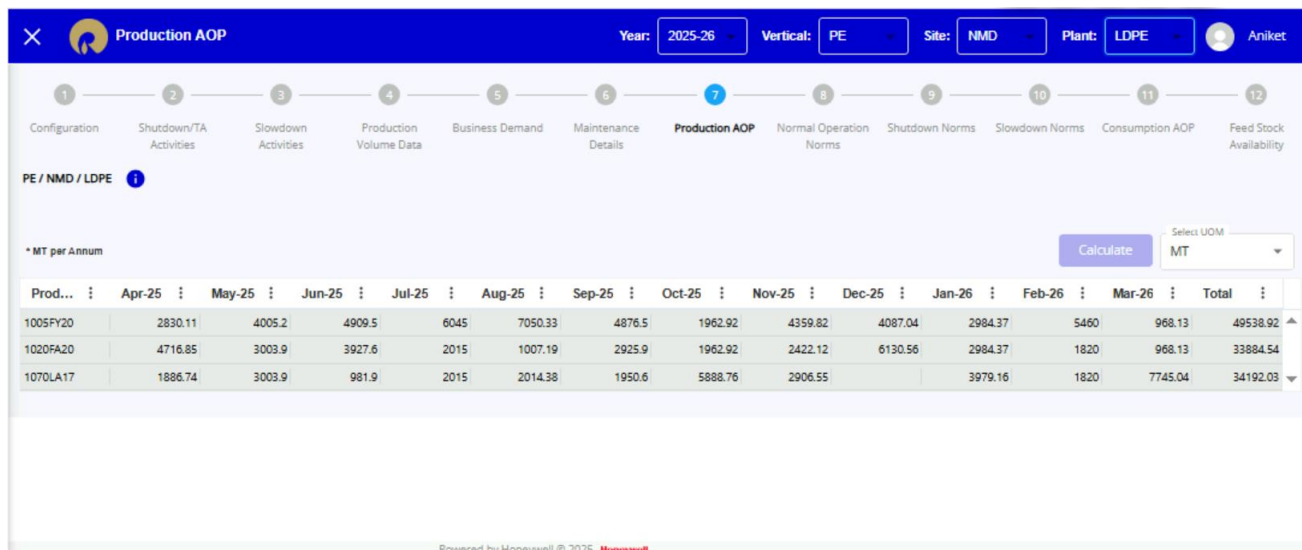
$$\begin{aligned} \text{Total Budgeted Monthly Production} \\ &= \text{Average MCU of All grades captured in section 2.1.1.2(TPH)} \\ &\quad * \text{Operating hours in a month (hr)} \end{aligned}$$

$$\begin{aligned} \text{Each Grade Production for a month} &= (\text{Total Budgeted Monthly Production} * \\ &\quad \text{Percentage of Grade production demand (\%)}) / 100 \end{aligned}$$

Calendar based production calculation explained above will be used to calculate monthly production of both LDPE and LLDPE1&2.

Hence, Month – wise production would be,

The screenshot below captures the calculated budgeted production values.



Prod...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
1005FY20	2830.11	4005.2	4909.5	6045	7050.33	4876.5	1962.92	4359.82	4087.04	2984.37	5460	968.13	49538.92
1020FA20	4716.85	3003.9	3927.6	2015	1007.19	2925.9	1962.92	2422.12	6130.56	2984.37	1820	968.13	33884.54
1070LA17	1886.74	3003.9	981.9	2015	2014.38	1950.6	5888.76	2906.55		3979.16	1820	7745.04	34192.03

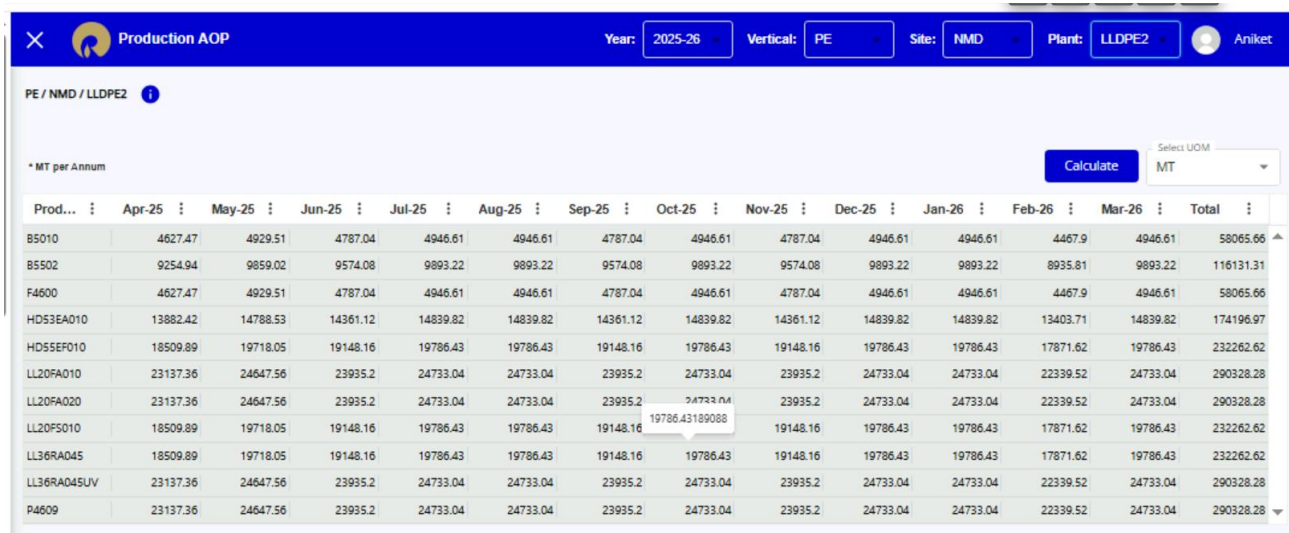
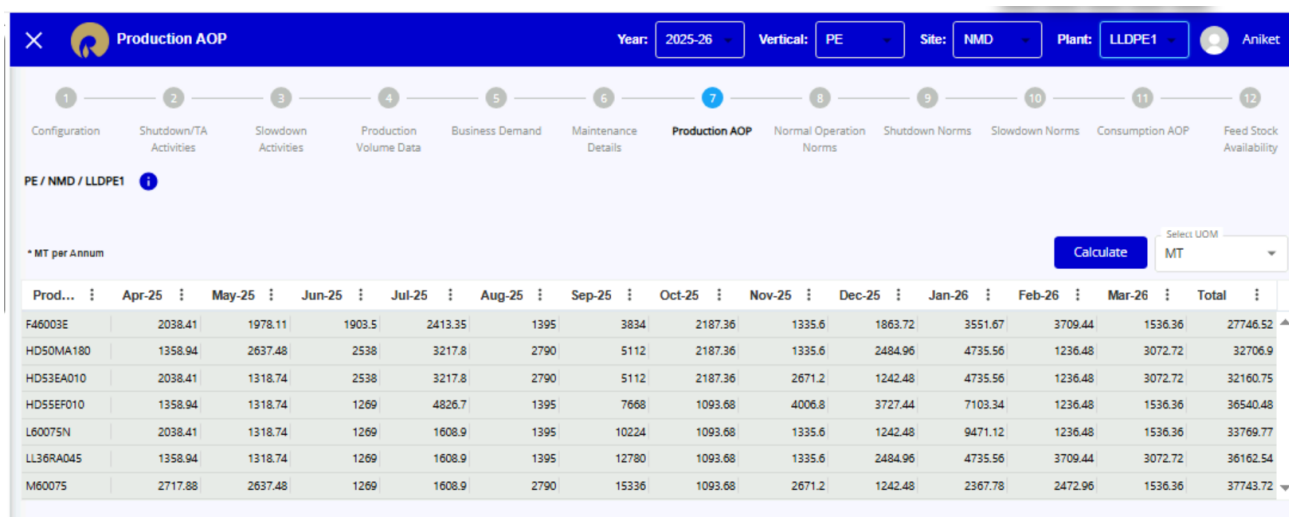


Figure 9 Grade wise Budget Production

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady-state Operations

At a high level, the norms calculation are categorized as below –

1. Best achieved –

Based on the Selected basis AOP Period corresponding to the MCU derived in section 2.1.1.2. High and Low range of MCU is calculated. Based on derived high and low range, production MIIS data for

selected period is filtered for consecutive days production. Consecutive days' selection is preferably 5 days, or if not available, 4, 3, 2, or 1 day. the days in which production lies within the high and low range and have consecutive days production are selected for norms calculation. Norms are calculated for each material and each day from the filtered data. respective material norms are then averaged out for Final norm value.

For each plant, conditions may vary as per information provided by Site planner. Below plant under NMD PE have condition for data filtering.

LDPE:

Condition1: 3% of MCU (derived in section 2.1.1.2) is used to find high low range of production. if any grade is missing in 3% of high & low range logic, it will try to increase percentage by +1 till 10% until all grades are covered.

Condition2: 5 Consecutive days production which all under high & Low range of production. The filtering for consecutive days is dynamic if no consecutive day's production is found for 5 logics will try to find for 4,3,2 or 1

Coverage: For LDPE Raw material, byproduct & utilities norms are generally covered under this calculation basis

LLDPE:

Condition1: 10% of MCU (derived in section 2.1.1.2) is used to find a high & low range of production.

Condition2: 5 Consecutive days production which all under high & Low range of production. The filtering for consecutive days is dynamic if no consecutive day's production is found for 5 logics will try to find for 4,3,2 or 1

Coverage: For LDPE Raw material, byproduct, utilities & catalyst-chemicals norms are generally covered under this calculation basis.

Coverage Exception: for utilities of LLDPE since consumption for each train is not available in any system additional data massaging is imposed after getting combined consumption from MIIS. Below is the calculation used to proportionate total consumption into each train consumption.

Actual daily utility consumption	Total Daily utility consumption as per norm	TR-1 utility consumption as per norm (Last year-based Norms)	TR-2 utility consumption as per norm (Last year-based Norms)	Diff btw Actual – norm (Last year-based Norms)	Tr-1 norm factor	TR-1 utility consumption after apportioning	TR-2 utility consumption after apportioning
A	B	C	D	E	F	G	H
Reconciled PAS value (LLDPE1_U TC001)	C + D	(Daily Tr-1 powder production from MCU * Grade wise last yr Budget Tr-1 utility Norm)	(Daily Tr-2 powder production from MCU * grade wise last yr Budget Tr-2 utility Norm)	A - B	C / B	C + (E*F)	A - G

Table 3 LLDPE Train1&2 Utilities Consumption Apportioning

Below is the formulated generalized logic for deriving the Best Achieved Norms. On top of below explained logic condition can be imposed which are different plant to plant basis as explained above

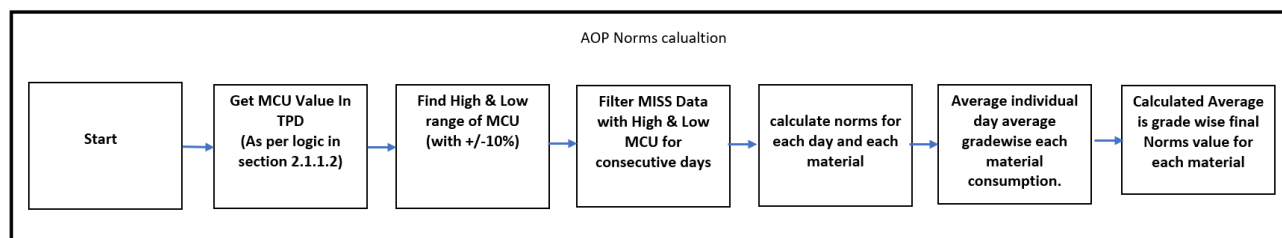


Figure 10 Norm Calculation Flow-Best Achieved

2. SAP Process order basis –

LDPE catalyst-chemicals norms are generally covered under this calculation basis. Based on the Selected basis AOP Period, SAP data for each process order is filter for catalyst & chemicals for each grade. Process order wise consumption aggregation is calculated. For the same process order Production is aggregated to calculate final production number for the process order. Further each material norm is calculated for each process order using the formula below.

$$\begin{aligned}
 & \text{Norms(per processorder)} \\
 &= \text{Aggregated Consumption(per processorder)} \\
 & \quad / \text{Aggregated Production (per processorder)}
 \end{aligned}$$

With the above logic applied on AOP period basis data we will get an array of data sets for each material for each grade per process order wise, now it is time to find final norm from the available data points. For specific to LDPE catalyst-chemicals logic used to calculate final norm is average out all norms after removing outlier to remove outlier logic used is explained below.

Outlier Removal:

Let's assume x, y, z are norms data set calculated with above explained logic.

Step 1 (A): Find average from all data points for each material (per grade).

Step 2 (B): Find standard deviation from all data points for each material (per grade).

Step 3(xi,yi,zi): calculate delta using norm data point average and standard deviation using below formula.

$$xi = Abs\left(\frac{x - A}{B}\right), yi = Abs\left(\frac{y - A}{B}\right), zi = Abs\left(\frac{z - A}{B}\right)$$

Step 4: removing data point (x,y,z...) (outlier removal) with condition remove norm data point if xi,yi,zi,...>1.

Step 5: average remaining norms data point which pass above condition.

The below screenshots depict the best-achieved feed stale-wise value captured in the AOP digitized platform

Normal Operation Norms

Year: 2025-26

Vertical: PE

Site: NMD

Plant: LDPE

Aniket

Select Grade

1020FA20

Save

Calculate

Part...	UO...	Apr...	May...	Jun...	Jul-25	Aug...	Sep...	Oct-25	Nov...	Dec...	Jan...	Feb...	Mar...	Re...
Raw Material														
ETHYLENE	MT	4963.53499	3160.99996	4133.00825	2120.38181	1059.86469	3078.92067	2065.57809	2548.7989	6451.18016	3140.44857	1915.18357	1018.7619	Click to add ...
By Products														
LDPE Offgas...	MT	30.62493	19.50332	25.5006	13.08272	6.53935	18.99689	12.74458	15.72605	39.80368	19.37652	11.81665	6.28574	Click to add ...
Utility Consumption														
Power_Dis	KWH	2851902.853...	1816218.649...	2374706.344...	1218309.720...	608967.42665	1769058.273...	1186821.099...	1464465.725	3706660.5046	1804410.416...	1100408.780...	585350.96083	Click to add ...
HP Steam_Dis	MT	597.64612	380.60764	497.64459	255.30956	127.6155	370.72469	248.71079	306.89413	776.76954	378.1331	230.60219	122.66643	Click to add ...
LP Steam_Dis	MT	258.93119	164.8989	215.60536	110.6133	55.28963	160.6171	107.75417	132.9624	336.53671	163.8268	99.90879	53.14544	Click to add ...
Cooling Wat...	M3	1355.5547	863.27756	1128.73563	579.08195	289.45189	840.8	720.2798579903284	696.08381	1761.83458	857.66492	523.04176	278.2266	Click to add ...
D M Water	M3	1730.81535	1102.26024	1441.20554	739.39025	369.58137	1073.63868	720.27986	888.78194	2249.56643	1095.09384	667.83636	355.24858	Click to add ...
Water	M3	268.85468	171.21863	223.8684	114.85254	57.4086	166.77272	111.88404	138.05816	349.43442	170.10544	103.73777	55.18223	Click to add ...
Nitrogen Gas	NM3	148606.32037	94639.11795	123740.67068	63483.41231	31731.9394	92181.69554	61842.61023	76310.05469	193145.8422	94023.81717	57339.85628	30501.33784	Click to add ...
COMPRESSE...	NM3	186871.1596	119007.8704	155602.82068	79829.84064	39902.63871	115917.68303	77766.54625	95959.23224	242879.22218	118234.13501	72104.37219	38355.16792	Click to add ...

Figure 11 Digitized AOP screen for Best Achieved Norms

3. Recipe Basis –

LLDPE catalyst-chemicals norms are generally covered under this calculation basis. catalyst consumption is calculated using standard recipe table and actual production for the day. standard recipe table can be found in DCT. The recipe table is also available for manual editing in AOP portal as shown in below screen capture.

Figure 12 Digitized AOP Recipe Screen

2.1.2.2 Shutdown & Slowdown Consumption.

Shutdown Consumption is calculated differently for LDPE and LLDPE.

LDPE

Shutdown consumption is calculated by shutdown hours which is calculated based on shutdown schedule and shutdown consumption per hour which either constants or manual entry

$$\text{Shutdown Consumption} = \text{Number of shutdown hours} * \text{Shutdown consumption rate}$$

LLDPE

Shutdown consumption is different for different material based on formula for each material. Formula involves constant values and number of shutdowns. Below screen capture represent shutdown & slowdown consumption calculation formula for ethylene accordingly each material have different formula mentioned in DCT.

Material	UoM	Number of continous GCO	Continuous GCO slowdown consumption	Number of discontinous GCO	Discontinuous GCO slowdown consumption	number of shutdown	shutdown consumption	Other losses	Monthly Total production from AOP	Grade wise - Steady state norm from AOP	Total loss
Ethylene	MT	a	5.25*a	b	10.248*b	c	10.248*c				5.25*a + 10.248*b + 10.248*c + Other losses

Figure 13 Material Consumption Formula for Shutdown & Slowdown

2.1.2.3 Budget Norms/ Consumption AOP

The shutdown consumption quantities will be summed with the steady-state operation norms, to derive the budgeted norms which will be inclusive of shutdown conditions. The intent here is to derive budgeted norms for the steady-state and shutdown months separately, using the logic below–

LDPE

Consumption Norm

$$= \text{Steady State Norms as captured in section 2.1.2.1} \\ + ((\text{Number of hours plant shutdown planned} * \text{Shutdown consumption per hour}) \\ + \text{Startuploss} + \text{Other losses}) / \text{Total Production})$$

$$\text{Consumption AOP} = \text{Consumption Norm} * \text{Grade Production AOP}$$

LLDPE1&2

Consumption Norm

$$= \text{Steady State Norms as captured in section 2.1.2.1} \\ + (\text{Continuous GCO slowdown consumption} \\ + \text{Discontinuous slowdown consumption} + \text{shutdown consumption} + \text{Other losses}) \\ / \text{Total Production})$$

$$\text{Consumption AOP} = \text{Consumption Norm} * \text{Grade Production AOP}$$

If shutdown consumption is not applicable for any month, its value will be 0. Hence, the above formula is applicable for all the months.

Consumption AOP

Year:

2025-26

Vertical:

PE

Site:

NMD

Plant:

LDPE

Aniket

PE / NMD / LDPE

Select Grade

1070LA17

Calculate

Parti...	UOM...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Raw Material													
ETHYLENE	MT	2026.00856	3182.88921	1040.42067	2134.65633	2133.79981	2066.26404	6243.92854	3082.8066		4217.88287	1927.47346	8203.11838
By Products													
MIXED OIL	MT	91.73136	146.04653	47.73897	97.96723	97.93708	94.83617	286.30545	141.31347		193.46267	88.48653	376.55587
LDPE Offgas t...	MT	96.5124	153.65848	50.22713	103.07329	103.04157	99.77903	301.22772	148.67874		203.54596	93.09845	396.182
Utility Consumption													
Power_Dis	KWH	1275486.40848	1995005.51784	652117.62526	1338237.7793	1337825.71081	1295467.15335	3910948.80707	1930350.92852		2642713.97065	1208731.68453	5143778.78155
HP Steam_Dis	MT	157.42758	214.00914	69.65181	143.63135	143.14742	139.60834	418.70478	206.84147		283.58047	129.51477	551.08914
LP Steam_Dis	MT	114.16991	167.50339	55.11818	113.85047	112.01639	109.57085	325.16398	160.20274		220.90662	100.33268	426.49072
Cooling Wate...	M3	28021.23971	881.51925	287.95441	592.33343	590.51173	571.76044	1726.02278	851.93998		1166.72775	533.66697	2274.15191
D M Water	M3	613.98691	934.23926	304.35567	624.46596	624.47397	604.92383	1826.81653	901.39381		1232.78879	564.28216	2399.53317
Water	M3	94.40578	136.42334	44.6992	91.92833	92.88036	88.80287	269.7561	132.60132		180.72297	84.31275	353.90365
Nitroen Gas	NM3	62178.63494	95476.1674	31208.2112	64042.67953	64023.5743	61996.29493	187164.76069	92377.92936		126471.94355	57844.81699	246160.94141

Figure 14 Digitized AOP screen for Month-wise Final Consumption AOP

2.1.3 Feedstock Availability

This section to be updated after discussion in detail with RIL on Site wide feedstock/AOP Production screen.

3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by NMD Site team